Ravi Dhaliwal

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Education

University of California, Los Angeles - Physics B.S. Dec 2025
Coursework: Graduate Plasma, Acoustics Laboratory, Statistical Mechanics, Analog Circuits
GPA: 3.4

Research Experience

Applied Machine Learning, Research Intern, General Atomics, DIII-D June 2026 - Aug 2026

- Using pytorch to do fast gaussian fitting on spectrometer output to determine plasma temperature and velocity
- Further characterizing plasma parameters in the transistion from low to high confinement, with data to be fed into a machine learning model for active control

Student Researcher, Putterman Group June 2025 – Present

- Worked with a graduate student on iteratively implementing a bubble collapse algorithm from a naval research paper, specifically a 1-D compressible viscous fluid lagrangian solver, implemented grid refinement, adaptive timestepping, and parallelized it to speed up computation by 8x.

Student Researcher, UCLA Basic Plasma Science Facility June 2024 – Present

- Analyzed how various beam characteristics and perturbation geometry affect DBS/CPS backscattered power.
- Developed a Full-Wave COMSOL model to simulate Doppler Backscattering and Cross-Polarization scattering in the DIII-D tokamak, confirming an intermediate regime of nonlinearity in the power response.
- Automated simulations for millimeter-wave propagation in plasma using GENRAY, integrating Python scripts with OMFIT's DBS module to streamline workflow for batch simulations and further analysis.

Aerodynamics Subteam, Bruin Formula SAE Sept 2023 – June 2024

- Mounted pitot tubes on custom 3D-printed fixtures to measure local airflow around the chassis.
- Ran STAR-CCM CFD studies and used results alongside measurements to influence drag-reduction design iterations.

Principal Investigator, NASA NPWEE – Remote Sept 2023 – Dec 2023

- Led a proposal for funding a conceptual engineering project to enhance thrust in satellite electronic propulsion systems.
- Specifically focused on investigating the feasibility of utilizing super-capacitors to induce large current increases in the thruster, which in turn increased thrust output
- Led research efforts, organized and managed teams by assigning roles and responsibilities, and conducted various simulations using pSpice
- Cumulated in a ten page proposal that was presented to a NASA board.

Student Researcher, University of California, Riverside – Riverside, CA June 2022 – Feb 2023

- Tasked with designing a venturi tube that would be used in a water filtration system
- Conducted CFD simulations in SolidWorks to verify the designs functionality under load, ensuring a sufficient pressure drop was achieved.

Publications

2-D Full-Wave Modeling of Doppler Backscattering from Coherent Plasma Density Fluctuations In Review

Projects

NAR Level Two Rocket Sept 2025

- Designed, Simulated, and Fabricated a 3 ft rocket with custom avionics, reaching an apogee of 1500m.

Skills

Software: COMSOL, STAR CCM+, SolidWorks, Inventor, AutoCAD
Languages: Python,MATLAB, R, SQL, C++, C, C#, Java